COMP 3059 – Capstone Project I

Software Requirements Analysis and Design Assignment

This assignment is an overview to gather the software needs with requirements analysis and help to proceed with the design.

The requirements analysis helps to break down functional and non-functional requirements to a basic design view to provide a clear system development process framework. It involves various entities, including business, stakeholders and technology requirements.

The design is the activity following requirements specification and before programming. Software design usually involves problem solving and planning a software solution.

To work on this assignment you could use the references and a sample template given below. The sample template can be customised to suit the nature of your project.

Reference Readings/Example to work on this Sprint document:
http://www.uacg.bg/filebank/acadstaff/userfiles/publ_bg_397_SDP_activities_and_steps.pdf

www.cse.msu.edu/~chengb/RE-491/Papers/SRSExample-webapp.doc

Source for this template:
www.tricity.wsu.edu/~mckinnon/cpts322/cpts322-srs-v1.doc

1.0 Introduction

1.1 Purpose

This document outlines the high-level software requirements for the *Hairstylist Booking Application*. It describes what the system will do, not how it will do it. The purpose is to ensure a shared understanding between the development team (*BrazWebDes*), the client (*Erick Henrique Lima Fernandes*), and stakeholders about the expected functionality, constraints, and dependencies of the system.

It serves as a foundation for:

- Guiding the system design and development.
- Providing a reference for testing and validation.
- Supporting future enhancements and maintenance.

Intended Audience:

- Project sponsor and client (Erick Henrique)
- Capstone instructor and evaluators
- BrazWebDes development team members

1.2 Scope

The *Hairstylist Booking Application* is a web and mobile solution that allows clients to schedule, reschedule, and cancel appointments online while providing the hairstylist with administrative tools to manage bookings, services, and availability.

In Scope

- Client booking flow (book, reschedule, cancel)
- Email confirmations and reminders
- Admin dashboard (manage services, hours, blackout dates, testimonials, blog posts)
- Portfolio & Testimonials module with consent registry
- Branding & Visual identity
- Mobile app prototype mirroring the client booking flow

Out of Scope

- Online payment integration
- Multi-staff or multi-location support (future enhancement)
- Advanced analytics and reporting
- Public commenting or community features

Goals and Benefits

- Replace manual DM/phone scheduling with a digital booking system.
- Reduce no-shows and scheduling conflicts.
- Enhance the hairstylist's online visibility and brand trust.
- Improve client experience with self-service booking and reminders.

2.0 System Overview

This is a new self-contained system, not a follow-on or replacement system. It is developed by the *BrazWebDes* team in collaboration with the client, *Erick Henrique Lima Fernandes*. The project integrates standard web and mobile technologies (HTML, CSS, JavaScript, responsive frameworks, email APIs) and is built from scratch based on requirements defined in earlier phases (Vision, HLR, and Project Plan).

2.1 Project Perspective

This is a new self-contained system, not a follow-on or replacement system. It is developed by the *BrazWebDes* team in collaboration with the client, *Erick Henrique Lima Fernandes*. The project integrates standard web and mobile technologies (HTML, CSS, JavaScript, responsive frameworks, email APIs) and is built from scratch based on requirements defined in earlier phases (Vision, HLR, and Project Plan).

2.2 System Context

The system operates within a small-business context, addressing the hairstylist's need for automation and professional branding.

Primary Entities

- **Client (User):** Interacts through a web or mobile interface to browse services, schedule, cancel, or reschedule appointments.
- **Admin (Erick):** Manages appointments, services, hours, and content.
- **Email Service:** Sends automatic confirmations, updates, and reminders.
- **Hosting Environment:** Deploys and stores the web app, database, and media content.

Strategic Considerations

- Supports digital transformation of small businesses.
- Enhances customer accessibility and experience.
- Enables future scalability for multi-staff or multi-location support.

2.3 General Constraints

- Single stylist, single location.

- Notifications via email only (no SMS).
- Hosting and email services must use free or low-cost tiers.
- Application must comply with privacy requirements (consent before publishing photos/testimonials).
- Must be developed and delivered within the Capstone course timeline (Sept 2025 – Mar 2026).

2.4 Assumptions and Dependencies

Assumptions

- The hairstylist (client) provides branding assets, photos, and testimonials.
- The system will run on standard browsers and modern mobile devices.
- The project will use shared GitHub repositories for collaboration.
- Internet connectivity is available for all users.

Dependencies

- Reliable hosting and transactional email provider.
- Timely feedback and approval from the client (Erick).
- Active participation of all team members (*BrazWebDes*).
- Integration between front-end and back-end components.

3.0 Functional Requirements

This section describes specific features of the Hairstylist Booking Application. Each functional requirement defines what the system will do to meet user and business needs. These requirements are based on the approved Project Vision, High-Level Requirements, and User Stories developed by Team BrazWebDes.

3.1 Functional Requirements

3.1.1 Functional Requirement #1 – Online Booking System

- **Introduction**
  The Online Booking System enables customers to view available services and time slots, book or reschedule appointments, and receive automated confirmation and reminder emails. This feature replaces manual DM or phone bookings, helping to reduce no-shows and scheduling conflicts.
- **Inputs**
  - Service selection (e.g., haircut, beard trim, eyebrows)
  - Preferred date and time
  - Customer details (name, email, phone number)

        - Optional notes or special requests
    - **Processing**
        - Validate selected service, date, and time.
        - Check availability based on the hairstylist's working hours and blackout dates.
        - Create a booking record in the database with a unique booking ID.
        - Trigger email confirmation and reminder notifications for both customer and hairstylist.
    - **Outputs**
        - Confirmation screen showing booking details.
        - Email confirmation and reminder messages sent to customer and hairstylist.
        - Updated booking calendar reflecting new appointment.

3.1.2 Functional Requirement #2 – Admin Dashboard

- **Introduction**
  The Admin Dashboard allows the hairstylist (Erick) to manage all operational aspects of the business, including appointments, services, client records, testimonials, and blog posts.
- **Inputs**
    - Admin login credentials
    - Service data (name, price, duration)
    - Availability parameters (working hours, blackout dates)
    - Booking actions (approve, reschedule, cancel)
- **Processing**
    - Authenticate admin credentials and validate permissions.
    - Store or update service, schedule, and booking data in the database.
    - Generate booking summaries and analytics.
- **Outputs**
    - Updated schedule and booking information on the dashboard.
    - Confirmation or alert messages upon successful updates.
    - Downloadable booking and service reports.

3.1.3 Functional Requirement #3 – Customer Profile System

- **Introduction**
  The Customer Profile System provides registered users with personalized access to their booking history, profile information, and preferences. It simplifies the rebooking process and improves user experience.
- **Inputs**
    - Login credentials

        - Profile information (email, phone, preferences)
        - Rebooking selections
    - **Processing**
        - Authenticate user credentials.
        - Retrieve and display customer-specific data.
        - Save any profile updates or new bookings to the database.
    - **Outputs**
        - Personalized dashboard with booking history and options to rebook.
        - Confirmation of updated profile data.

3.1.4 Functional Requirement #4 – Portfolio and Testimonials

- **Introduction**
  This module allows customers to submit testimonials and optional photos, which the admin can approve before publishing. It builds credibility and showcases the hairstylist's previous work.
- **Inputs**
    - Customer feedback and optional image upload.
    - Admin approval or rejection.
- **Processing**
    - Capture explicit consent from the customer for displaying their photo or testimonial.
    - Store content in the database for review.
    - Publish approved testimonials and portfolio items on the website.
- **Outputs**
    - Publicly visible testimonials and portfolio gallery.
    - Secure database record of consent.
    - Admin confirmation of successful publication.

3.1.5 Functional Requirement #5 – Blog Module (Optional)

- **Introduction**
  The Blog Module allows the hairstylist to create, edit, and publish articles or announcements, improving SEO and engagement.
- **Inputs**
    - Blog post title, content, cover image, tags, and status (draft/publish).
- **Processing**
    - Validate content and metadata.
    - Store blog entries in the database.
    - Make posts visible to the public upon publication.
- **Outputs**

- Blog section displaying published posts.
- Admin confirmation of successful creation or update.

3.1.6 Functional Requirement #6 – Notification Service

- **Introduction**
  The Notification Service automates communication through email confirmations, reminders, and updates to both customers and the admin.
- **Inputs**
  - Event triggers (new booking, reschedule, cancellation).
  - Customer and admin email addresses.
- **Processing**
  - Detect relevant event and generate corresponding email template.
  - Send messages through the configured email service provider.
  - Log notification details for auditing.
- **Outputs**
  - Confirmation and reminder emails.
  - Notification logs accessible to admin.

3.2 Use Cases

Use Cases describe how users (actors) interact with the system to achieve specific goals.
Actors identified for this system include:

- **Customer (Client)** – interacts with the web/mobile application.
- **Admin (Hairstylist – Erick)** – manages system content and bookings.
- **Email Service** – external system for transactional email delivery.

3.2.1 Use Case #1 – Book Appointment

- **Actor:** Customer
  Goal: Schedule a service appointment online.
- **Preconditions:**
  - Customer has access to the website or mobile app.
  - Service list and time slots are available.
- **Main Flow:**
  - Customer selects a service.
  - System displays available time slots.
  - Customer chooses date and time and enters contact details.

        - System validates inputs and creates a booking.
        - Confirmation page and email are sent to both parties.
    - **Alternative Flow:**
        - If selected time slot is unavailable, the system shows alternative options.
    - **Postconditions:**
        - Booking data stored in the database and displayed on admin calendar.

3.2.2 Use Case #2 – Manage Appointments

- **Actor:** Admin (Hairstylist)
  Goal: Approve, modify, or cancel bookings.
- **Preconditions:**
    - Admin is logged into the dashboard.
- **Main Flow:**
    - Admin opens schedule view.
    - Selects a booking.
    - Updates, reschedules, or cancels appointment.
    - System saves the change and notifies the client.
- **Postconditions:**
    - Booking table and admin dashboard reflect changes.

3.2.3 Use Case #3 – Submit Testimonial

- **Actor:** Customer
  Goal: Provide feedback and photo for publication.
- **Preconditions:**
    - Customer has completed a booking.
- **Main Flow:**
    - Customer submits testimonial and optional image.
    - System requests consent for publishing.
    - Admin reviews and approves the submission.
    - Testimonial appears publicly upon approval.
- **Postconditions:**
    - Testimonial and consent record saved in database.

3.2.4 Use Case #4 – Publish Blog Post

- **Actor:** Admin (Hairstylist)
  Goal: Create and publish a new blog post.
- **Preconditions:**
    - Admin logged into dashboard.

- **Main Flow:**
  - Admin creates a blog post (title, content, image).
  - Adds SEO tags and metadata.
  - Publishes the post.
- **Postconditions:**
  - Blog post becomes visible on the public website.

3.2.5 Use Case #5 – Receive Email Notifications

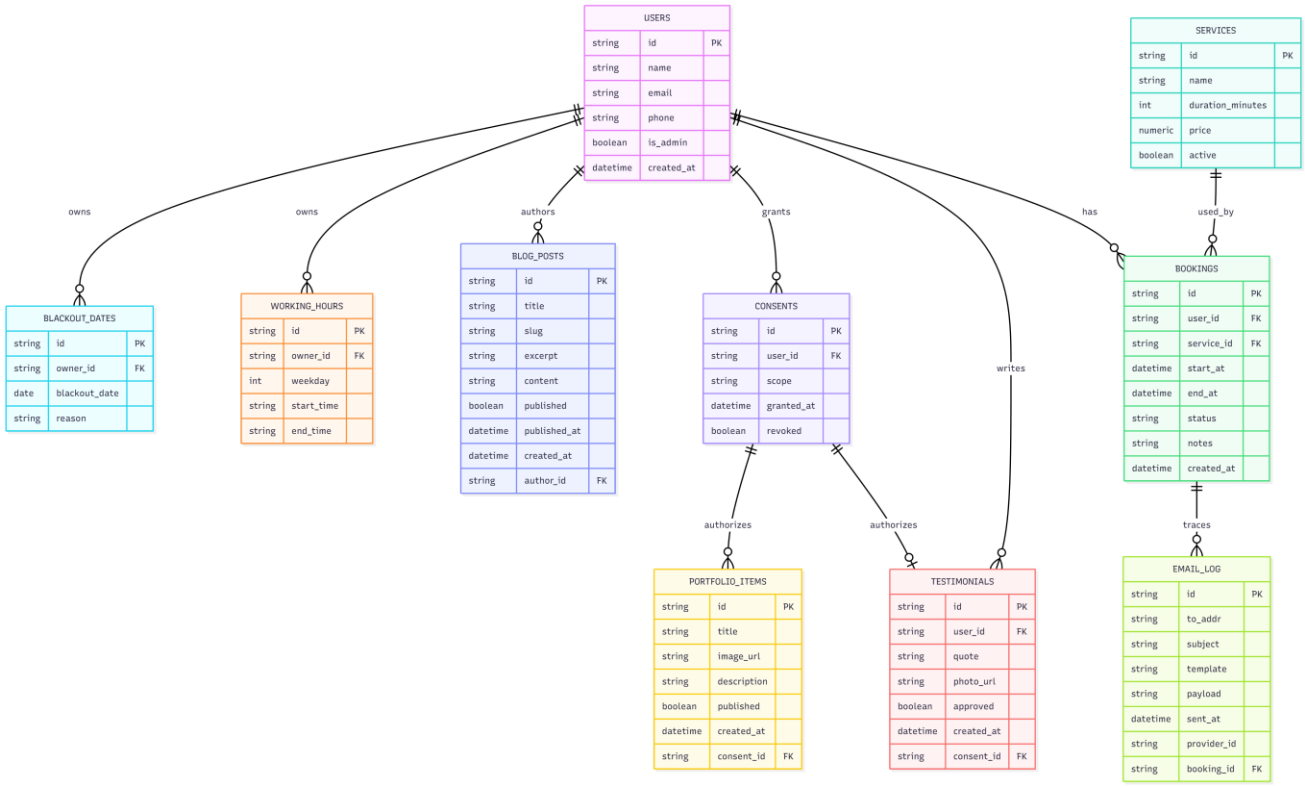
- **Actors:** Customer, Admin, Email Service
  Goal: Automatically send booking notifications.
- **Preconditions:**
  - Booking, reschedule, or cancellation event occurs.
- **Main Flow:**
  - System detects booking event.
  - Generates appropriate email template.
  - Sends message via the Email Service.
  - Logs notification in the system.
- **Postconditions:**
  - Emails delivered to recipients and logged for verification.

3.3 Data Modelling and Analysis

3.3.1 Normalized Data Model Diagram (ERD)

This ERD captures the core entities and relationships used by the application: users, services, bookings, availability (working hours and blackout dates), content (testimonials, portfolio items, blog posts), consent records, and email logs. The model supports the booking lifecycle (create, reschedule, cancel), content publishing with consent, and system notifications.

3.3-1: Normalized ERD

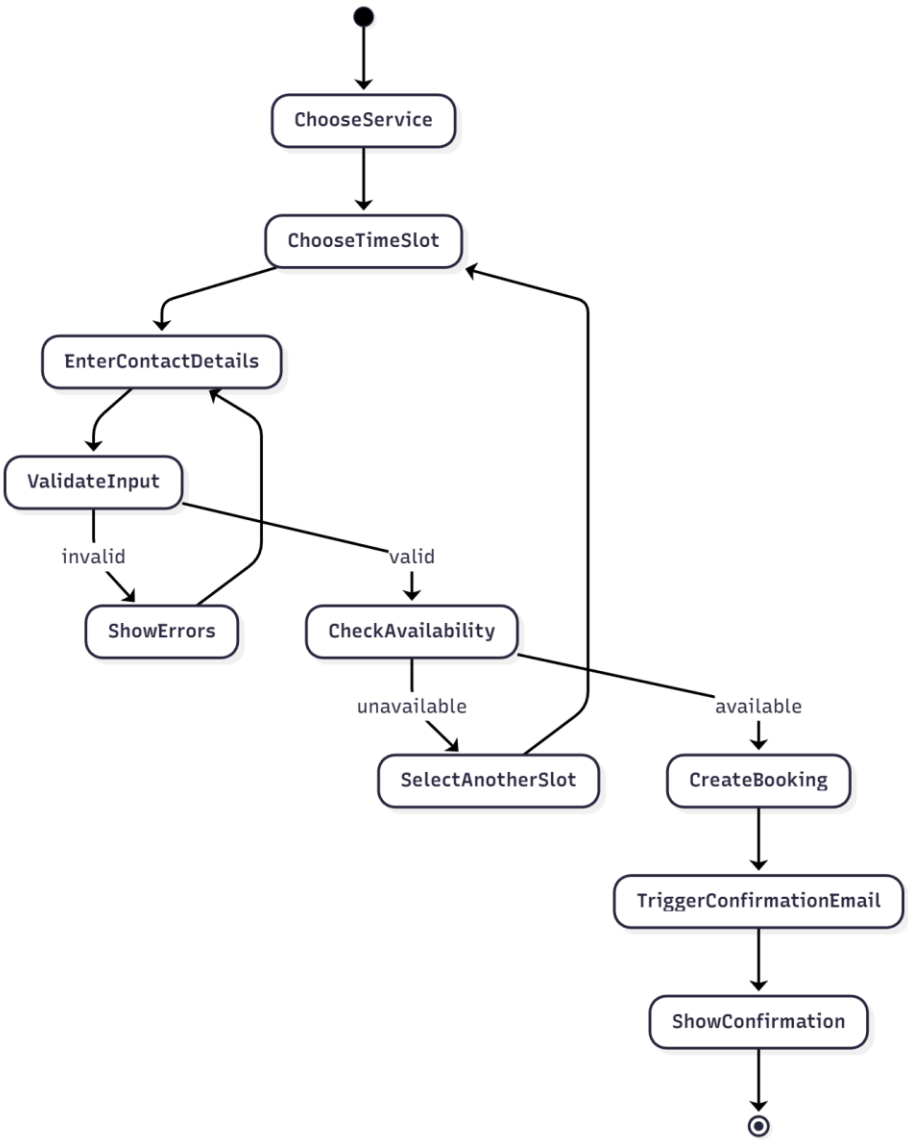


3.3.2 Activity Diagrams

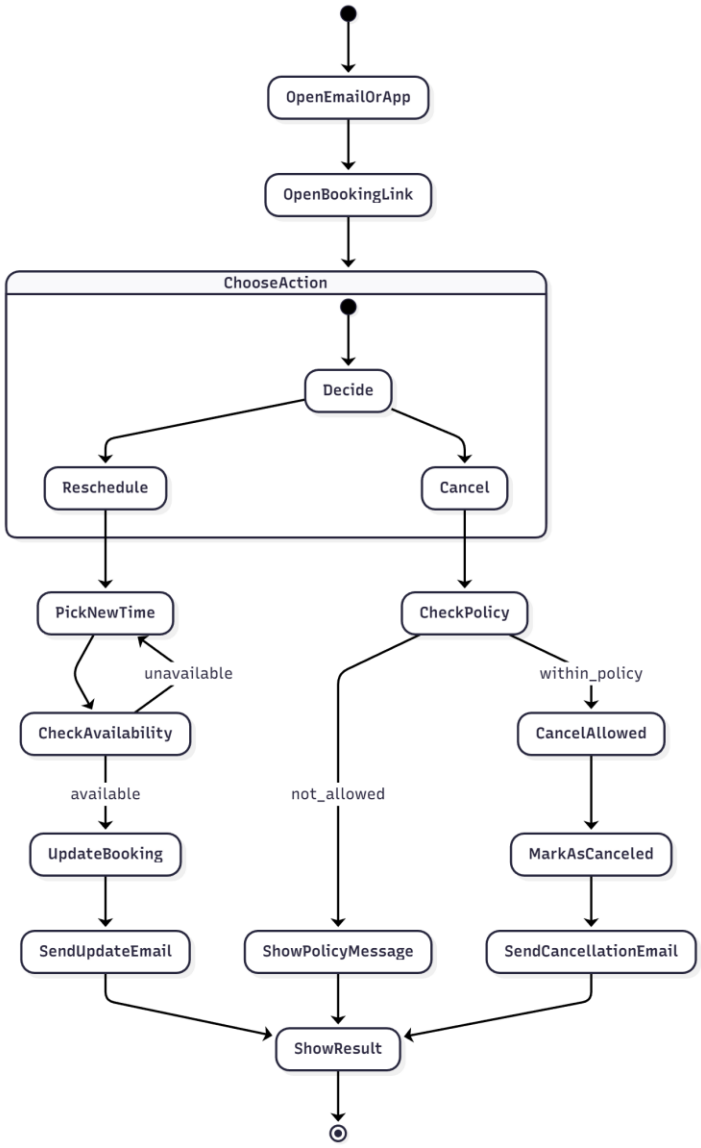
The following activity diagrams depict key user and admin flows. Together, they describe how customers book and manage appointments and how admins configure availability and publish content.

- AD-1 Booking a Service (Customer)
- AD-2 Reschedule or Cancel (Customer)
- AD-3 Manage Services and Hours (Admin)
- AD-4 Create & Publish Blog Post (Admin)
- AD-5 Add Portfolio Item (Admin) with Consent Check

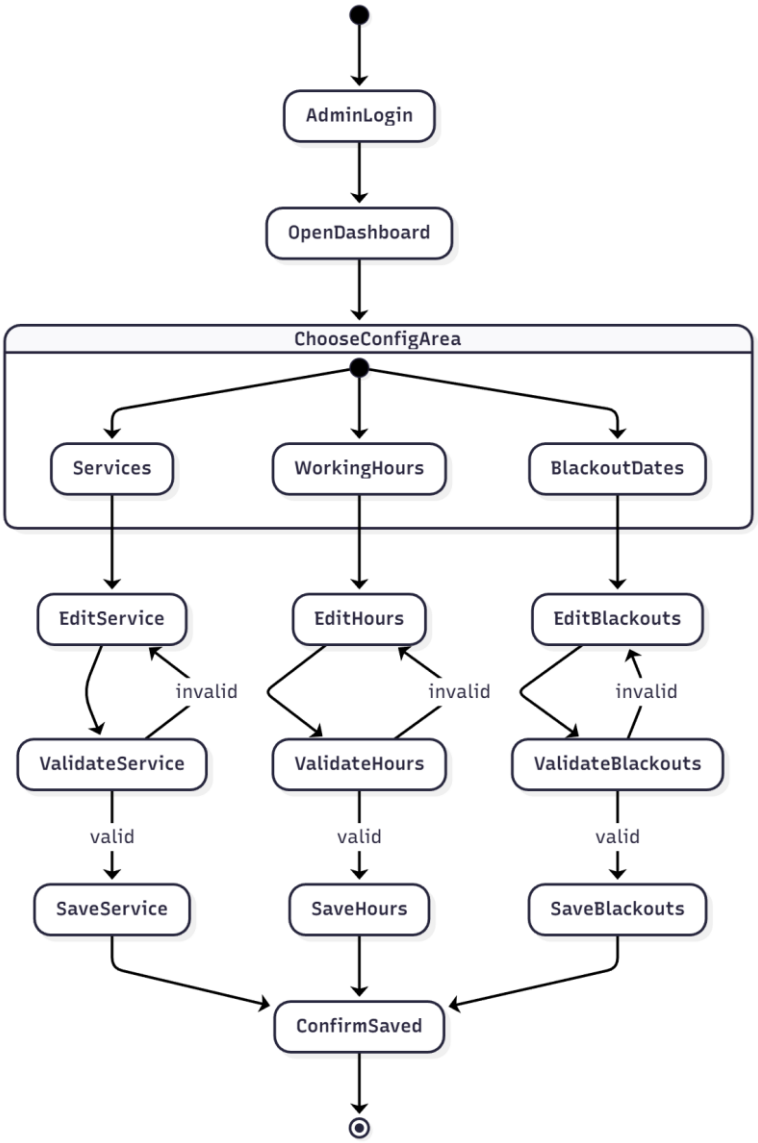
3.3-2A: Activity — Booking a Service (Customer)



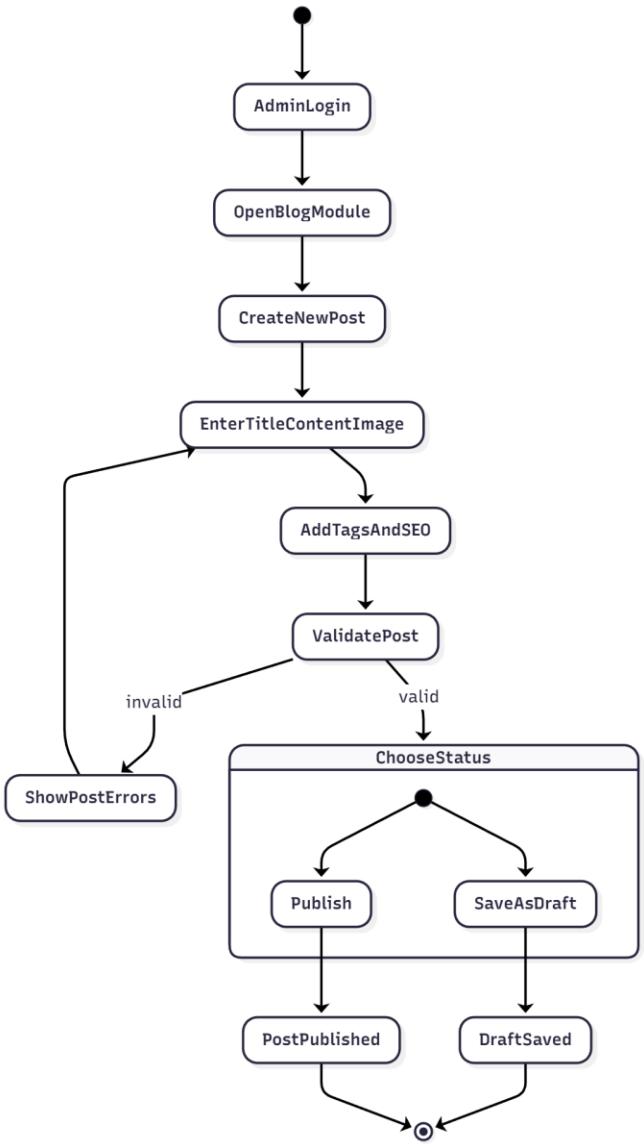
3.3-2B: Activity — Reschedule or Cancel (Customer)



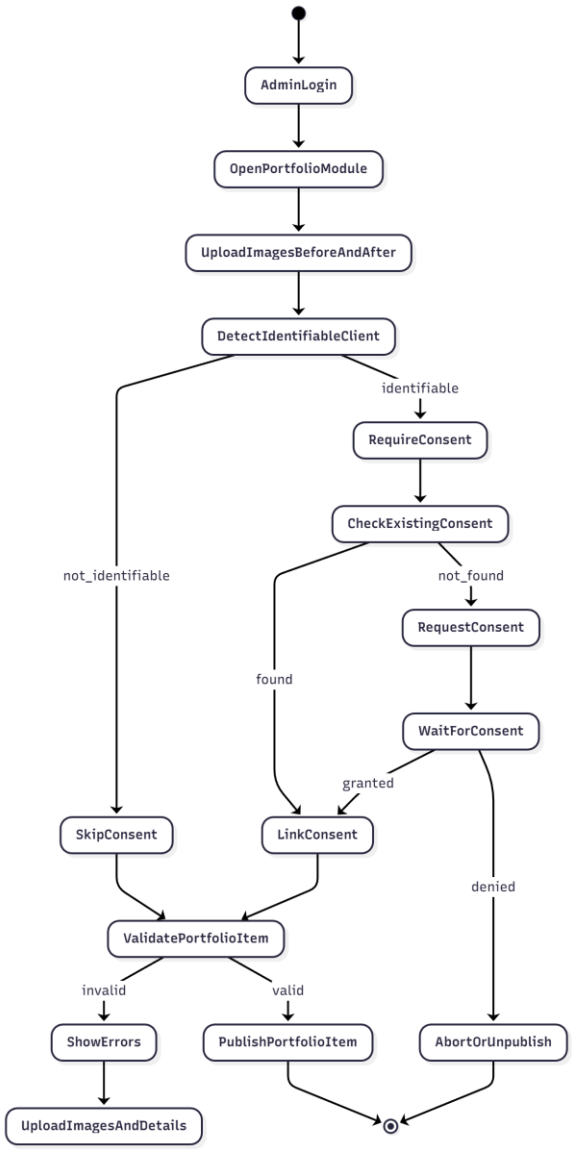
3.3-2C: Activity — Manage Services and Hours (Admin)



3.3-2D: Activity — Create & Publish Blog Post (Admin)



3.3-2E: Activity — Add Portfolio Item (Admin) with Consent Check



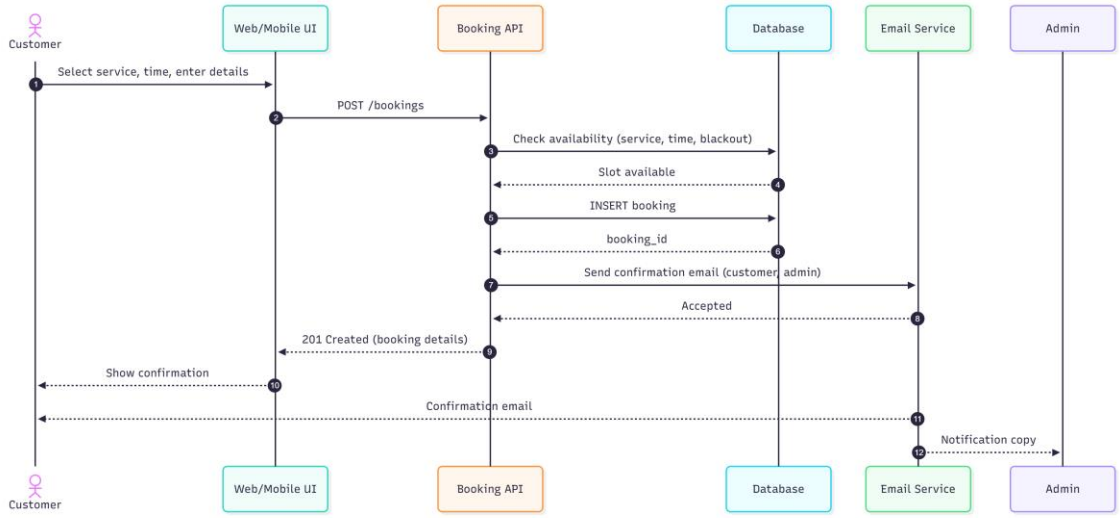
3.3.3 Sequence Diagrams

The sequence diagrams detail message exchanges among UI, APIs, database, email service, and admin actors across the main scenarios.

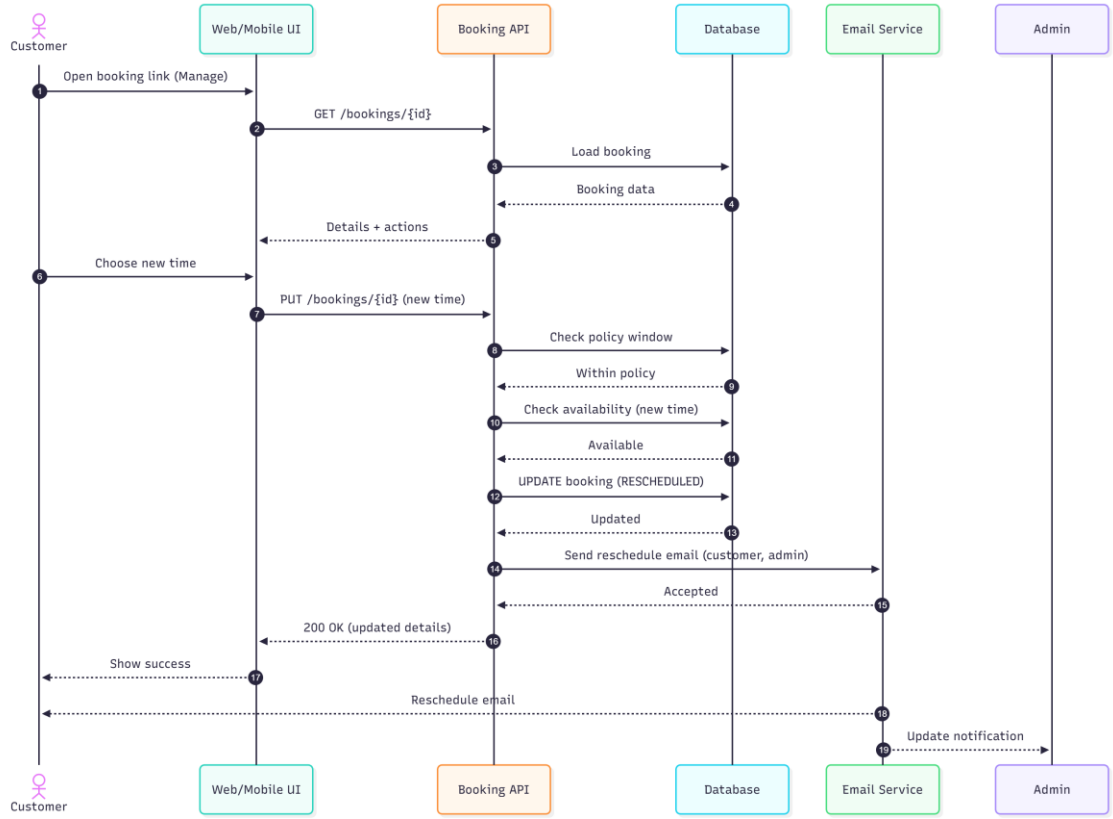
- SD-1 Book Appointment with Email Confirmation
- SD-2 Manage Appointment (Reschedule and Cancel)
- SD-3 Add Portfolio Item (Admin)

- SD-4 Manage Services and Hours (Admin)

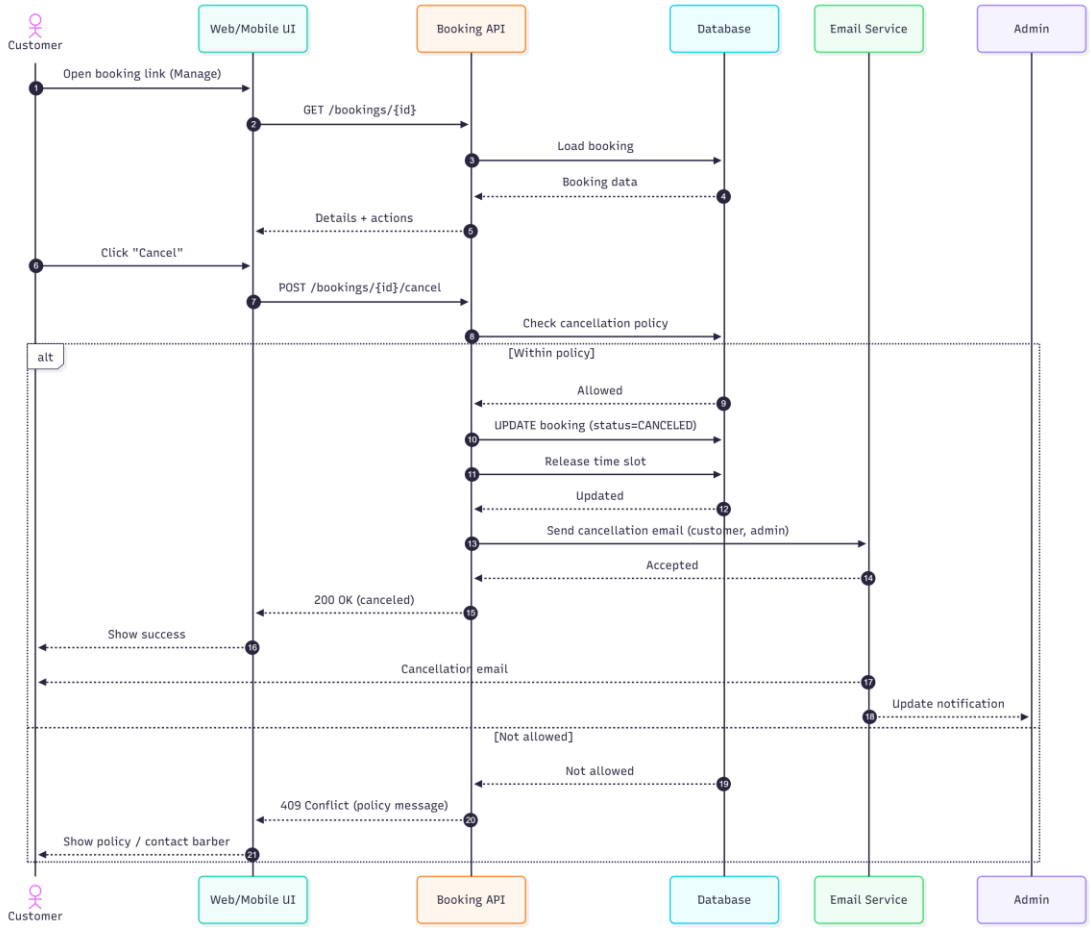
3.3-3A: Sequence — Book Appointment (Confirmation Email)



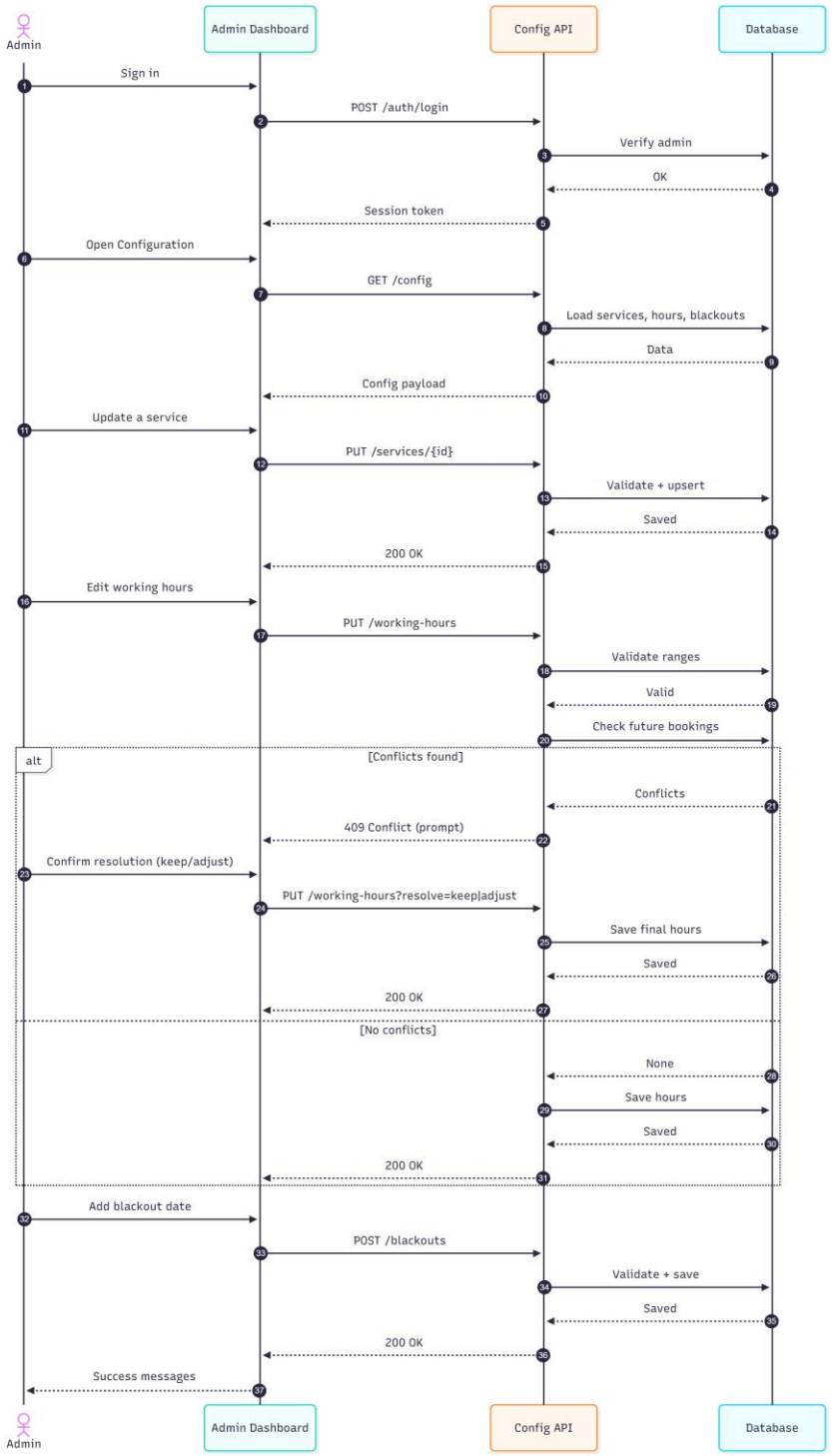
3.3-3B: Sequence — Manage Appointment (Reschedule and Cancel)



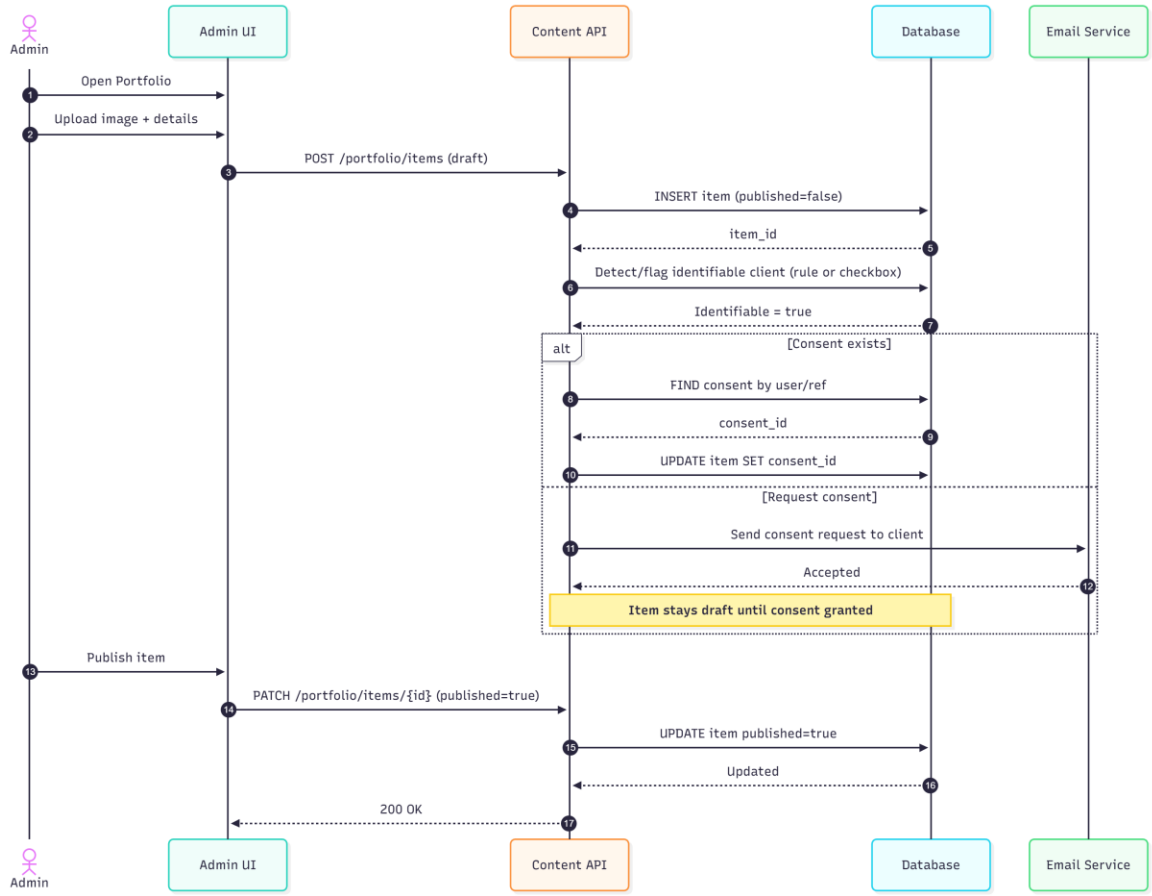
Cancel:



3.3-3C: Sequence — Add Portfolio Item (Admin)



Insert Figure 3.3-3D: Sequence — Manage Services and Hours (Admin)

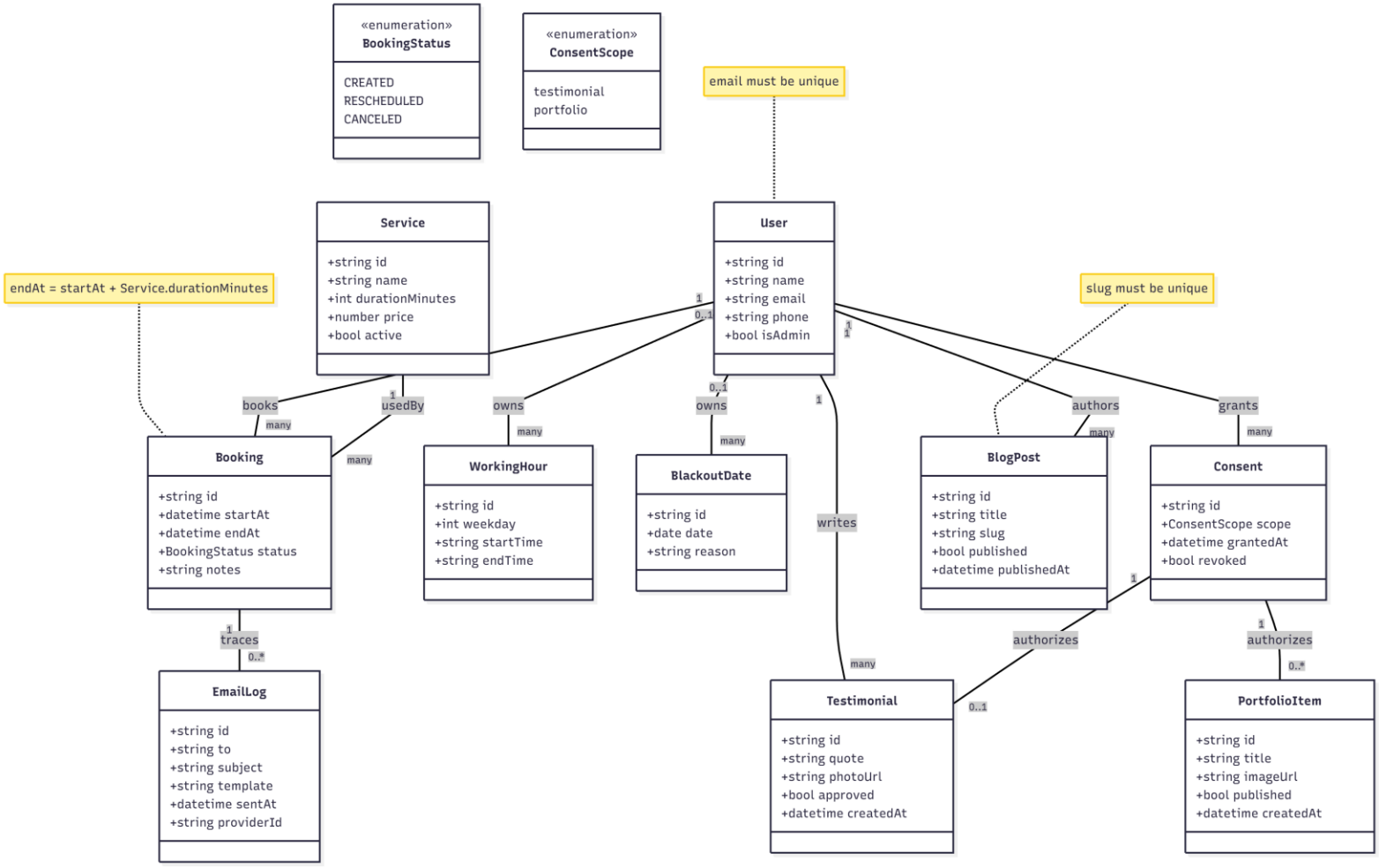


3.3.4 UML Class Diagram

UML Class Diagram — (Domain + Application Services)

This class diagram provides the domain model used by the system: User, Service, Booking (with BookingStatus), WorkingHour, BlackoutDate, Consent (with ConsentScope), Testimonial, PortfolioItem, BlogPost, and EmailLog, including the main associations.

3.3-4: UML Class Diagram

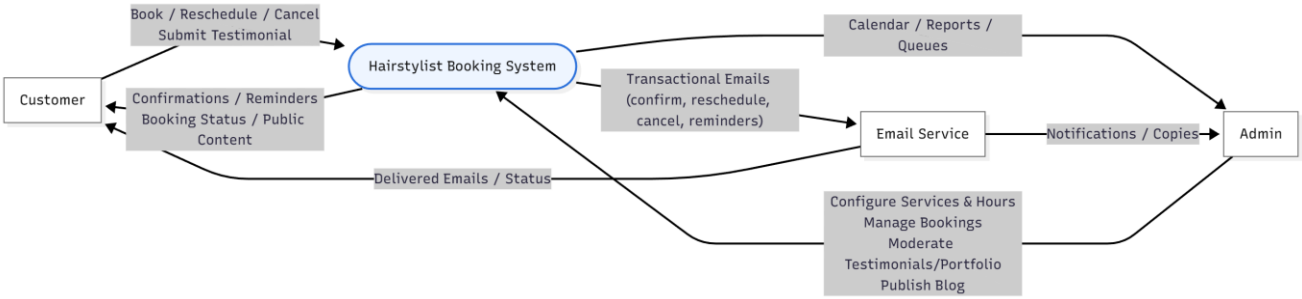


3.4 Process Modelling

3.4.1 Data Flow Diagram — Level 0 (Context)

The Level-0 DFD shows the system as a single process interacting with external entities (Customer, Admin, Email Service). It highlights the major information exchanges: booking/reschedule/cancel requests, testimonials and content actions, administrative configuration, and transactional emails.

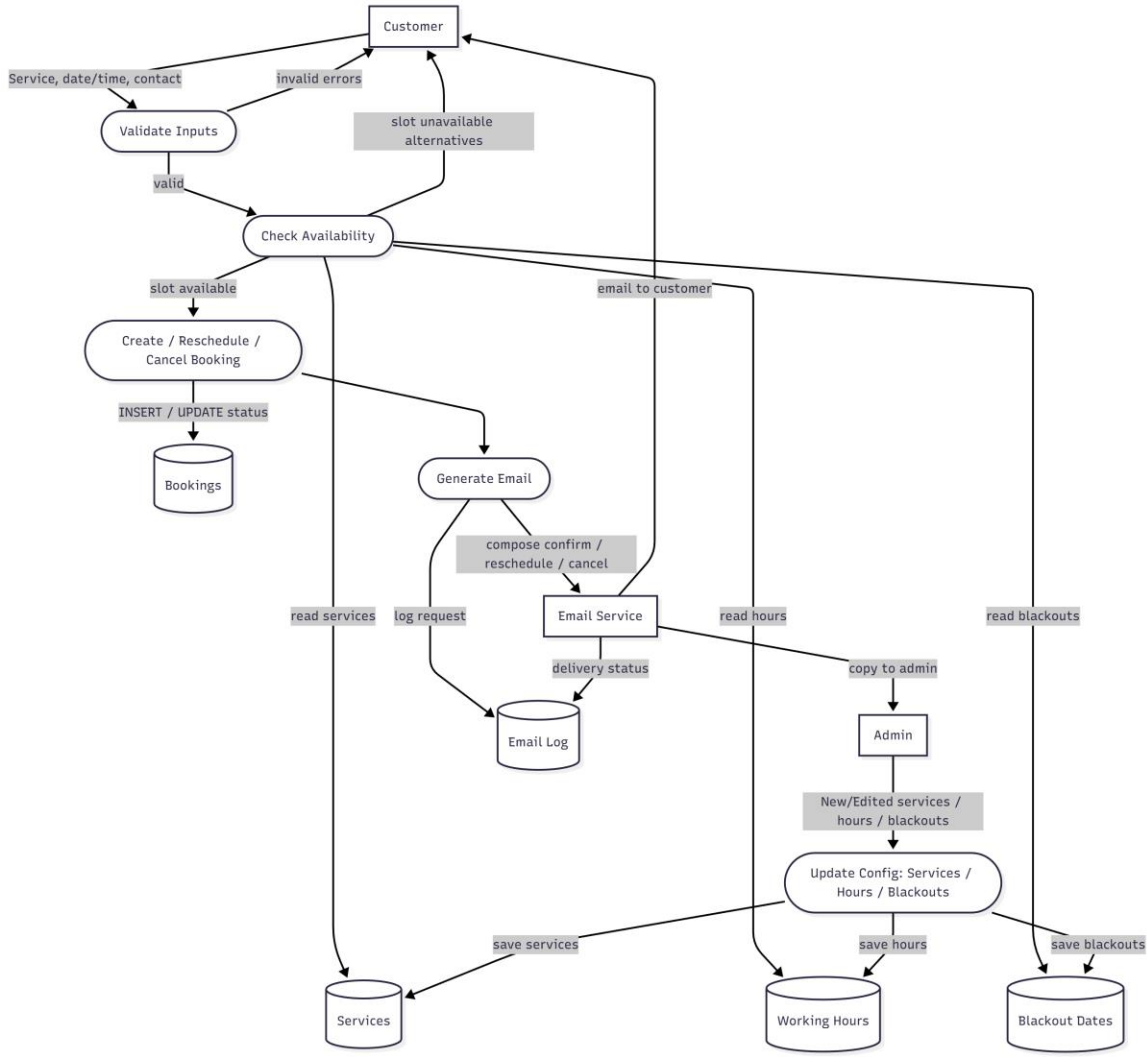
3.4-1: DFD Level 0 (Context)



3.4.2 Data Flow Diagram — Level 1 (Booking + Reschedule/Cancel + Admin Impact)

The Level-1 DFD expands the booking process into validation, availability checks (services, working hours, blackout dates), create/reschedule/cancel operations, and email generation with logging. It also shows how admin configuration (services/hours/blackouts) updates the data stores that drive availability.

3.4-2: DFD Level 1 — Booking & Notifications (with Admin Configuration impact)



4.0 Non-Functional Requirements

4.1 Performance

- **NFR-01 – Page / Screen Load Time**
  - 95% of public pages (Home, Services, Booking, Portfolio, Blog) should load in under 3 seconds on a typical home or mobile connection.
- **NFR-02 – Booking Response Time**
  - After the user submits a booking form with valid data, the system should confirm or reject the request in under 2 seconds, under normal load.
- **NFR-03 – Concurrent Usage**
  - The system must support at least 10 concurrent users (browsing, booking, or managing the calendar) without noticeable degradation in response time.

4.2 Reliability & Availability

- **NFR-04 – Uptime Target**
  - The application should be available at least 95% of the time during Erick's working hours (Toronto time), excluding scheduled maintenance.
- **NFR-05 – Error Handling**
  - In case of failures (e.g., email provider issues, DB errors), the system must show a clear, user-friendly message and log the error for later review, without exposing technical details.
- **NFR-06 – Data Persistence**
  - Confirmed bookings, customer records, and testimonials must be stored in a persistent database so that data is not lost after system restarts or deployments.

4.3 Security & Privacy

- **NFR-07 – Secure Transport**
  - All traffic for the web application must use HTTPS, ensuring encryption of personal data (names, emails, phone numbers).
- **NFR-08 – Authentication & Authorization**
  - Admin features (calendar management, services, testimonials approval, blog management) must be accessible only to authenticated admin users.
- **NFR-09 – Data Protection**
  - Personal information and booking data must be stored securely, with access limited to authorized processes and users.
- **NFR-10 – Consent for Media**
  - Any client photo or testimonial shown in the Portfolio & Testimonials section must have recorded consent and be removable upon request.

4.4 Usability & Accessibility

- **NFR-11 – Mobile-Friendly Design**
  - The web app must be responsive, usable on phones, tablets, and desktops without horizontal scrolling for main flows.
- **NFR-12 – Clear Booking Flow**
  - The booking process should be completed in no more than 4 main steps (service → date/time → personal info → review/confirm).
- **NFR-13 – Accessibility**
  - The UI should follow basic accessibility practices (legible font sizes, sufficient color contrast, focus indicators, and labels on form fields).

4.5 Maintainability

- **NFR-14 – Code Organization**
  - The system should follow a modular structure (separate layers for front-end, back-end, and data access) to make it easier to update parts without breaking others.
- **NFR-15 – Documentation**
  - Core features (booking, admin dashboard, email notifications, portfolio, blog) must be documented with setup steps and basic maintenance instructions so another developer can take over if needed.

4.6 Portability

- **NFR-16 – Deployment Environment**
  - The web application should be deployable on common cloud environments (e.g., Azure) with minimal configuration changes.
- **NFR-17 – Mobile Application**
  - The Android mobile app should run on a commonly used Android version (e.g., Android 12 or later), on at least one test device defined by the team.

5.0 Logical Database Requirements

The system will use a relational database (PostgreSQL) to store and manage all key information related to:

- Users (clients and admin)
- Services
- Appointments
- Availability and blackout dates
- Portfolio photos and testimonials
- Blog posts
- Consent and notification logs

The database must ensure data integrity**,** consistency**,** and reliability.

Data Formats

- **Timestamps** will follow the ISO-8601 standard (e.g., 2025-11-05T12:30:00-05:00) to ensure consistency and proper handling of time zones.
- **Email addresses** will follow standard email formats and be stored as unique when used for login.

- **Phone numbers** (if provided) should be stored in a normalized format (e.g., E.164) to support future integrations (WhatsApp, SMS) even if not used right away.
- Text fields (e.g., testimonial message, blog content) will use Unicode-compatible types to support special characters and accents.

Data Retention

- **Appointment records** will be retained for several years (e.g., 3–7 years) to help with history and potential business reporting.
- **Notification logs** (email confirmations, reminders) will be kept for a shorter period (e.g., 90 days) mainly for troubleshooting and verification.
- **Portfolio items and testimonials** will remain stored until the owner deletes/unpublishes them or the client requests removal.
- **Soft deletion** will be used for key entities (e.g., users, testimonials) so records can be hidden from normal operations without being immediately removed from the database, supporting potential recovery and audit.

Integrity

- Tables will use primary keys and foreign keys to enforce referential integrity (e.g., each appointment must reference a valid user and service).
- Unique constraints will be enforced where appropriate (e.g., email in the users table) to avoid duplicate accounts.
- Transactional consistency will be enforced: for example, creating an appointment and writing the corresponding notification log should either both succeed or both fail, preventing partial/inconsistent records.

Backups and Recovery

- The database will have regular automated backups (at least once per day).
- Backups will be stored securely with access limited to authorized team members/admins.
- The goal is to support point-in-time recovery within approximately 24 hours in case of data loss or corruption.
- Periodic restore tests should be performed (during the project or as part of handover instructions) to verify that backups are usable.

6.0 Other Requirements

This section covers additional non-functional and regulatory aspects needed for a professional, maintainable, and compliant system.

Legal and Policy Compliance

- The system will provide clear Terms of Use, Privacy Policy, and Cancellation Policy, accessible from the website and surfaced during the booking flow (e.g., via links and/or a checkbox).
- The application will follow Canadian privacy principles (e.g., PIPEDA):
    - collect only necessary personal data (name, email, phone, booking details);
    - explain how data is used (booking, reminders, communication);
    - allow clients to request removal of their testimonials or photos.
- Any photo/testimonial displayed publicly must be linked to a recorded consent in the database, and there must be a way to unpublish/remove it on request.

Internationalization and Accessibility

- The first release will be in English only, but content and UI will be structured so that additional languages can be added later (e.g., text not hard-coded everywhere).
- The UI will follow responsive design and mobile-first principles, ensuring usability on phones, tablets, and desktops.
- The design will be guided by WCAG 2.1 AA accessibility principles where feasible, such as:
    - sufficient color contrast;
    - keyboard navigation;
    - descriptive alt text for key images (especially portfolio items that represent services).

Analytics and Monitoring

- The system may use basic**,** privacy-respecting analytics to understand usage patterns (e.g., page views, most visited services, booking funnel drop-off).
- Where analytics tools are used, they should:
    - anonymize IPs and avoid storing unnecessary personal data;
    - allow users to opt out of tracking (e.g., via a cookie/consent banner);
    - store analytics data separately from main user PII whenever possible.
- Basic error logging and uptime monitoring should be in place so the team (or future maintainer) can detect failures and fix them promptly.

7.0 Approval

The signatures below indicate their approval of the contents of this document.

Project Role	Name	Signature	Date
Project Manager	Renan Yoshida Avelan	Renan Yoshida Avelan	Nov-10-2025
Lead Developer	Gustavo Miranda	Gustavo Miranda	Nov-10-2025
Frontend Developer	Lucas Tavares Criscuolo	Lucas Tavares Criscuolo	Nov-10-2025
Backend Developer	Mateus Carvalho de Souza Melco Sfeir	Mateus Carvalho de Souza Melco Sfeir	Nov-10-2025